

Web Services CA 1

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API Screenshots

/getSingleProduct

GET **/getSingleProduct** Get Single Product

Return a single product by ID.

Parameters

Name	Description
id * required integer (query)	Product ID 1047

Response body

```
{
  "_id": "69ccb16403293f42cd36cdfc",
  "ProductID": 1047,
  "Name": "ASUS ProArt Z790",
  "UnitPrice": 470,
  "StockQuantity": 6,
  "Description": "Creator-focused motherboard with Thunderbolt 4."
}
```

/getAll

GET **/getAll** Get All

Return all products in the inventory.

Parameters

No parameters

Response body

```
[
  {
    "_id": "69ccb16403293f42cd36cdd0",
    "ProductID": 1001,
    "Name": "NVIDIA RTX 4090",
    "UnitPrice": 1599.99,
    "StockQuantity": 12,
    "Description": "High-end GPU with 24GB VRAM for 4K gaming."
  },
  {
    "_id": "69ccb16403293f42cd36cdd1",
    "ProductID": 1002,
    "Name": "AMD Ryzen 9 7950X",
    "UnitPrice": 549,
    "StockQuantity": 25,
    "Description": "16-core 32-thread enthusiast desktop processor."
  },
  {
    "_id": "69ccb16403293f42cd36cdd2",
    "ProductID": 1003,
    "Name": "Samsung 990 Pro 2TB",
    "UnitPrice": 179.99,
    "StockQuantity": 40,
    "Description": "PCIe Gen4 NVMe M.2 SSD with 7450MB/s reads."
  },
  {
    "_id": "69ccb16403293f42cd36cdd3",
    "ProductID": 1004,
```

/addNew

POST

/addNew Add New

Add a new product to the inventory.

Parameters

No parameters

Request body required

Edit Value | Schema

```
{
  "ProductID": 2001,
  "Name": "Logitech MX Master 3S",
  "UnitPrice": 50.00,
  "StockQuantity": 78,
  "Description": "Wireless mouse"
}
```

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/addNew' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "ProductID": 2001,
    "Name": "Logitech MX Master 3S",
    "UnitPrice": 50.00,
    "StockQuantity": 78,
    "Description": "Wireless mouse"
  }'
```

/deleteOne

DELETE**/deleteOne** Delete One

Delete a product by ID.

Parameters

Name	Description
id * required	Product ID to delete
integer (query)	2001

Response body

```
{
  "message": "Product deleted successfully"
}
```

/startswith

GET

/startsWith Starts With

Return all products whose name starts with the given letter (case-insensitive).

Parameters

Name	Description
letter * required	Letter to filter by

string
(query)

minlength: 1
maxlength: 1

Response body

```
[
  {
    "_id": "69ccb16403293f42cd36cdd0",
    "ProductID": 1001,
    "Name": "NVIDIA RTX 4090",
    "UnitPrice": 1599.99,
    "StockQuantity": 12,
    "Description": "High-end GPU with 24GB VRAM for 4K gaming."
  },
  {
    "_id": "69ccb16403293f42cd36cdd6",
    "ProductID": 1007,
    "Name": "NZXT H7 Flow Case",
    "UnitPrice": 129.99,
    "StockQuantity": 22,
    "Description": "Mid-tower ATX case with high-airflow mesh front."
  },
  {
    "_id": "69ccb16403293f42cd36cdd7",
    "ProductID": 1008,
    "Name": "Noctua NH-D15",
    "UnitPrice": 109.95,
    "StockQuantity": 45,
    "Description": "Dual-tower flagship CPU air cooler."
  },
  {
    "_id": "69ccb16403293f42cd36ce09",
    "ProductID": 1058,
```

/paginate

GET

/paginate Paginate

Return products in batches of 10 between start and end ID.

Parameters

Name	Description
------	-------------

startId * required	Starting product ID
---------------------------	---------------------

integer (query)	1050
--------------------	-----------------------------------

endId * required	Ending product ID
-------------------------	-------------------

integer (query)	1080
--------------------	-----------------------------------

Response body

```
[
  {
    "_id": "69ccb16403293f42cd36ce01",
    "ProductID": 1050,
    "Name": "Thermal Grizzly Kryonaut",
    "UnitPrice": 11.99,
    "StockQuantity": 150,
    "Description": "High-end thermal grease for overclocking."
  },
  {
    "_id": "69ccb16403293f42cd36ce02",
    "ProductID": 1051,
    "Name": "Sapphire Pulse RX 7800 XT",
    "UnitPrice": 499.99,
    "StockQuantity": 15,
    "Description": "Efficient 1440p gaming AMD GPU."
  },
  {
    "_id": "69ccb16403293f42cd36ce03",
    "ProductID": 1052,
    "Name": "Intel Core i5-12400F",
    "UnitPrice": 149,
    "StockQuantity": 45,
    "Description": "Best value budget gaming CPU (no iGPU)."
  },
  {
    "_id": "69ccb16403293f42cd36ce04",
    "ProductID": 1053,
```

/convert

GET **/convert** Convert

Return the product's price converted from USD to EUR using live exchange rate.

Parameters

Name	Description
id * required integer (query)	Product ID to get price in EUR

```
{
  "ProductID": 1123,
  "Name": "MSI MAG Z790 Tomahawk",
  "PriceUSD": 269.99,
  "PriceEUR": 234.82,
  "ExchangeRate": 0.86972
}
```

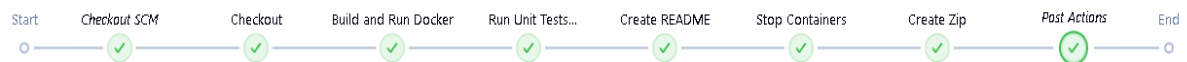
Jenkins Builds

Stages

April 1, 2026



Pipeline Overview for #4



End of Console Output

```
[Pipeline] }  
[Pipeline] // node  
[Pipeline] End of Pipeline  
Finished: SUCCESS
```

Zip file Created

Console output

```
[Pipeline] // stage  
[Pipeline] stage  
[Pipeline] { (Create Zip)  
[Pipeline] powershell  
Created: complete-2026-04-01-07-03-31.zip  
[Pipeline] }
```


Zip contents

complete-2026-04-01-07-03-31					Search complete-
Sort View					Details
Name	Date modified	Type	Size		
 docker-compose.yml	01/04/2026 06:47	YAML	2 KB		
 Dockerfile	01/04/2026 06:09	File	1 KB		
 Jenkinsfile	01/04/2026 07:01	File	5 KB		
 main.py	01/04/2026 06:09	Python Source File	7 KB		
 postman_collection.json	01/04/2026 06:09	JSON Source File	2 KB		
 postman_environment.json	01/04/2026 06:09	JSON Source File	1 KB		
 products.csv	01/04/2026 06:09	Microsoft Excel Co...	15 KB		
 prometheus.yml	01/04/2026 06:09	YAML	1 KB		
 README.txt	01/04/2026 07:03	Text Document	2 KB		

A-grade component

Prometheus table

 Prometheus

Query Alerts Status

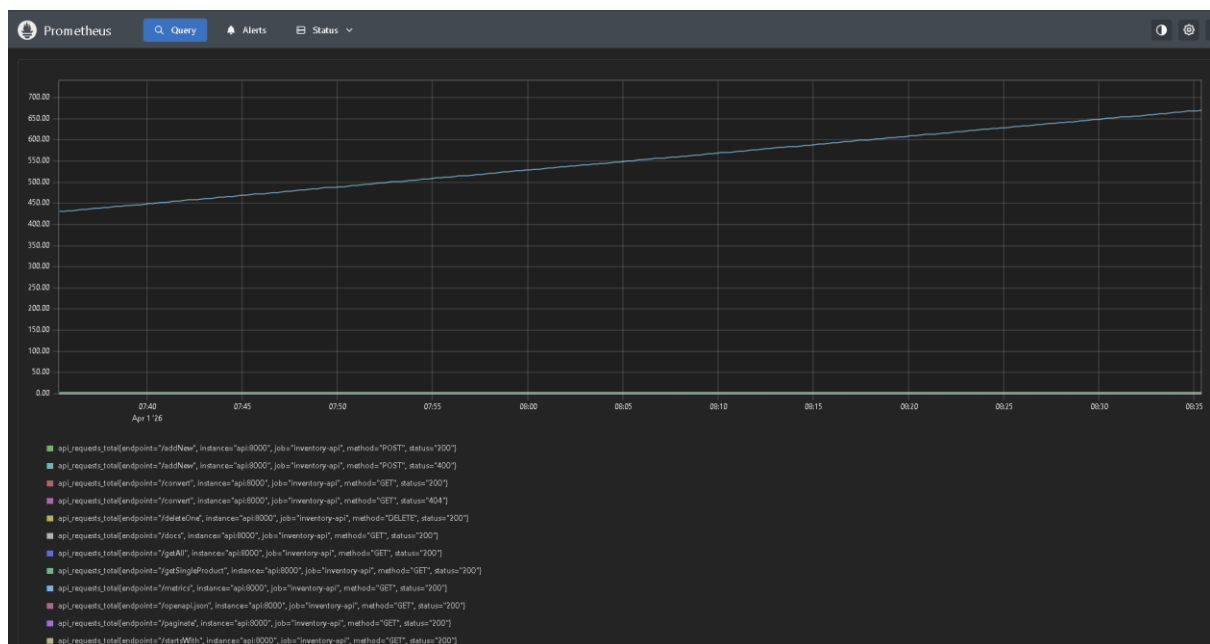
>_ api_requests_total

Table Graph Explain

< Evaluation time >

api_requests_total{endpoint="/metrics", instance="api:8000", job="inventory-api", method="GET", status="200"}
api_requests_total{endpoint="/docs", instance="api:8000", job="inventory-api", method="GET", status="200"}
api_requests_total{endpoint="/openapi.json", instance="api:8000", job="inventory-api", method="GET", status="200"}
api_requests_total{endpoint="/getSingleProduct", instance="api:8000", job="inventory-api", method="GET", status="200"}
api_requests_total{endpoint="/getAll", instance="api:8000", job="inventory-api", method="GET", status="200"}
api_requests_total{endpoint="/addNew", instance="api:8000", job="inventory-api", method="POST", status="200"}
api_requests_total{endpoint="/addNew", instance="api:8000", job="inventory-api", method="POST", status="400"}
api_requests_total{endpoint="/deleteOne", instance="api:8000", job="inventory-api", method="DELETE", status="200"}
api_requests_total{endpoint="/startsWith", instance="api:8000", job="inventory-api", method="GET", status="200"}
api_requests_total{endpoint="/paginate", instance="api:8000", job="inventory-api", method="GET", status="200"}
api_requests_total{endpoint="/convert", instance="api:8000", job="inventory-api", method="GET", status="404"}
api_requests_total{endpoint="/convert", instance="api:8000", job="inventory-api", method="GET", status="200"}

Prometheus graph



Metrics endpoint

```
localhost:8000/metrics

# HELP api_requests_total Total API requests
# TYPE api_requests_total counter
api_requests_total{endpoint="/metrics",method="GET",status="200"} 692.0
api_requests_total{endpoint="/docs",method="GET",status="200"} 1.0
api_requests_total{endpoint="/openapi.json",method="GET",status="200"} 1.0
api_requests_total{endpoint="/getSingleProduct",method="GET",status="200"} 4.0
api_requests_total{endpoint="/getAll",method="GET",status="200"} 1.0
api_requests_total{endpoint="/addNew",method="POST",status="200"} 1.0
api_requests_total{endpoint="/addNew",method="POST",status="400"} 1.0
api_requests_total{endpoint="/deleteOne",method="DELETE",status="200"} 1.0
api_requests_total{endpoint="/startsWith",method="GET",status="200"} 1.0
api_requests_total{endpoint="/paginate",method="GET",status="200"} 1.0
api_requests_total{endpoint="/convert",method="GET",status="404"} 1.0
api_requests_total{endpoint="/convert",method="GET",status="200"} 1.0
```

Appendix

Fast API Code

main.py:

import os

```

from contextlib import asynccontextmanager

import httpx

from fastapi import FastAPI, HTTPException, Query

from pydantic import BaseModel, Field

from pymongo import MongoClient

from prometheus_client import Counter, Histogram, generate_latest,
CONTENT_TYPE_LATEST

from fastapi.responses import Response


# MongoDB config
MONGODB_URI = os.getenv("MONGODB_URI", "mongodb://localhost:27017")
DATABASE_NAME = "inventory_db"
COLLECTION_NAME = "products"


# Prometheus metrics (API monitoring)
REQUEST_COUNT = Counter(
    "api_requests_total",
    "Total API requests",
    ["method", "endpoint", "status"]
)

REQUEST_LATENCY = Histogram(
    "api_request_duration_seconds",
    "API request latency in seconds",
    ["endpoint"]
)


# Pydantic models for validation
class Product(BaseModel):
    ProductID: int = Field(..., gt=0, description="Unique product identifier")

```

```
Name: str = Field(..., min_length=1, max_length=200)
```

```
UnitPrice: float = Field(..., ge=0)
```

```
StockQuantity: int = Field(..., ge=0)
```

```
Description: str = Field(..., min_length=1)
```

```
class ProductCreate(BaseModel):
```

```
    ProductID: int = Field(..., gt=0)
```

```
    Name: str = Field(..., min_length=1, max_length=200)
```

```
    UnitPrice: float = Field(..., ge=0)
```

```
    StockQuantity: int = Field(..., ge=0)
```

```
    Description: str = Field(..., min_length=1)
```

```
def get_db():
```

```
    client = MongoClient(MONGODB_URI)
```

```
    return client[DATABASE_NAME][COLLECTION_NAME]
```

```
@asynccontextmanager
```

```
async def lifespan(app: FastAPI):
```

```
    # Startup and check MongoDB connection
```

```
    try:
```

```
        get_db().find_one()
```

```
    except Exception as e:
```

```
        print(f"Warning: MongoDB connection failed: {e}")
```

```
    yield
```

```
    # Shutdown
```

```
    pass
```

```
app = FastAPI(
```

```
    title="Inventory Management API",
```

```
description="Complete API for product inventory management",
version="1.0.0",
lifespan=lifespan,
)
```

```
# Middleware for Prometheus metrics
```

```
@app.middleware("http")
```

```
async def metrics_middleware(request, call_next):
```

```
    import time
```

```
    start = time.time()
```

```
    response = await call_next(request)
```

```
    duration = time.time() - start
```

```
    REQUEST_COUNT.labels(
```

```
        method=request.method,
```

```
        endpoint=request.url.path,
```

```
        status=response.status_code
```

```
    ).inc()
```

```
    REQUEST_LATENCY.labels(endpoint=request.url.path).observe(duration)
```

```
    return response
```

```
@app.get("/getSingleProduct")
```

```
async def get_single_product(
```

```
    id: int = Query(..., gt=0, description="Product ID")
```

```
):
```

```
    """Return a single product by ID"""
```

```
    product = get_db().find_one({"ProductID": id})
```

```
    if not product:
```

```
        raise HTTPException(status_code=404, detail="Product not found")
```

```
    product["_id"] = str(product["_id"])
```

```
return product
```

```
@app.get("/getAll")
```

```
async def get_all():
```

```
    """Return all products in the inventory"""
```

```
    products = list(get_db().find({}))
```

```
    for p in products:
```

```
        p["_id"] = str(p["_id"])
```

```
    return products
```

```
@app.post("/addNew")
```

```
async def add_new(product: ProductCreate):
```

```
    """Add a new product to the inventory"""
```

```
    db = get_db()
```

```
    if db.find_one({"ProductID": product.ProductID}):
```

```
        raise HTTPException(status_code=400, detail="Product ID already exists")
```

```
    doc = product.model_dump()
```

```
    db.insert_one(doc)
```

```
    return {"message": "Product added successfully", "ProductID": product.ProductID}
```

```
@app.delete("/deleteOne")
```

```
async def delete_one(
```

```
    id: int = Query(..., gt=0, description="Product ID to delete")
```

```
):
```

```
    """Delete a product by ID"""
```

```
    result = get_db().delete_one({"ProductID": id})
```

```
    if result.deleted_count == 0:
```

```
        raise HTTPException(status_code=404, detail="Product not found")
```

```
    return {"message": "Product deleted successfully"}
```

```

@app.get("/startsWith")
async def starts_with(
    letter: str = Query(..., min_length=1, max_length=1, description="Letter to filter by")
):
    """Return all products whose name starts with the given letter"""
    letter = letter.upper()
    products = list(get_db().find({
        "Name": {"$regex": f"^{letter}", "$options": "i"}
    }))
    for p in products:
        p["_id"] = str(p["_id"])
    return products

```

```

@app.get("/paginate")
async def paginate(
    start_id: int = Query(..., gt=0, alias="startId", description="Starting product ID"),
    end_id: int = Query(..., gt=0, alias="endId", description="Ending product ID")
):
    """Return products in batches of 10 between start and end ID"""
    if start_id > end_id:
        raise HTTPException(status_code=400, detail="startId must be <= endId")
    products = list(get_db().find(
        {"ProductID": {"$gte": start_id, "$lte": end_id}}
    ).sort("ProductID", 1).limit(10))
    for p in products:
        p["_id"] = str(p["_id"])
    return products

```



```

@app.get("/convert")
async def convert(
    id: int = Query(..., gt=0, description="Product ID to get price in EUR")
):
    """Return the product's price converted from USD to EUR"""
    product = get_db().find_one({"ProductID": id})
    if not product:
        raise HTTPException(status_code=404, detail="Product not found")

    async with httpx.AsyncClient() as client:
        resp = await client.get(
            "https://api.frankfurter.dev/v1/latest?base=USD&symbols=EUR"
        )
        if resp.status_code != 200:
            raise HTTPException(status_code=502, detail="Exchange rate API unavailable")
        data = resp.json()
        rate = data["rates"]["EUR"]

    price_usd = product["UnitPrice"]
    price_eur = round(price_usd * rate, 2)
    return {
        "ProductID": id,
        "Name": product["Name"],
        "PriceUSD": price_usd,
        "PriceEUR": price_eur,
        "ExchangeRate": rate,
    }

# Prometheus metrics endpoint for API monitoring

```

```

@app.get("/metrics")
async def metrics():
    """Prometheus metrics for API performance monitoring"""
    return Response(
        content=generate_latest(),
        media_type=CONTENT_TYPE_LATEST
    )

if __name__ == "__main__":
    import uvicorn
    uvicorn.run(app, host="0.0.0.0", port=8000)

```

import_csv_to_mongodb.py:

```

import csv
import json
import os
from pymongo import MongoClient

# MongoDB connection
MONGODB_URI = os.getenv("MONGODB_URI", "mongodb://localhost:27017")
DATABASE_NAME = "inventory_db"
COLLECTION_NAME = "products"
CSV_FILE = "products.csv"

def convert_csv_to_json(csv_path: str) -> list[dict]:
    """Read CSV and convert to list of JSON-compatible dicts"""
    products = []
    with open(csv_path, "r", encoding="utf-8") as f:
        reader = csv.DictReader(f)

```

```

for row in reader:

    product = {
        "ProductID": int(row["ProductID"]),
        "Name": row["Name"],
        "UnitPrice": float(row["UnitPrice"]),
        "StockQuantity": int(row["StockQuantity"]),
        "Description": row["Description"],
    }

    products.append(product)

return products

```

```

def import_to_mongodb(products: list[dict]) -> None:
    """Insert products into MongoDB, replacing existing collection"""
    client = MongoClient(MONGODB_URI)
    db = client[DATABASE_NAME]
    collection = db[COLLECTION_NAME]

    # Clear existing data and insert fresh
    collection.delete_many({})
    collection.insert_many(products)

    # Create index on ProductID for fast lookups
    collection.create_index("ProductID", unique=True)

    print(f"Successfully imported {len(products)} products to MongoDB")
    client.close()

def main():
    script_dir = os.path.dirname(os.path.abspath(__file__))

```

```

csv_path = os.path.join(script_dir, CSV_FILE)

if not os.path.exists(csv_path):
    print(f"Error: {CSV_FILE} not found at {csv_path}")
    return

products = convert_csv_to_json(csv_path)
import_to_mongodb(products)

if __name__ == "__main__":
    main()

```

Unit Test Code

Postman_collection.json:

```

{
  "info": {
    "name": "Inventory API Tests",
    "schema": "https://schema.getpostman.com/json/collection/v2.1.0/collection.json"
  },
  "item": [
    {
      "name": "Test getSingleProduct",
      "event": [
        {
          "listen": "test",
          "script": {
            "type": "text/javascript",
            "exec": [

```

```

    "pm.test('Status code is 200', function () {",
    "  pm.response.to.have.status(200);",
    "});",
    "pm.test('Response is JSON', function () {",
    "  pm.response.to.be.json;",
    "});",
    "pm.test('Product has correct ID', function () {",
    "  var jsonData = pm.response.json();",
    "  pm.expect(jsonData.ProductID).to.eql(1001);",
    "});",
    "pm.test('Product has all required fields', function () {",
    "  var jsonData = pm.response.json();",
    "  pm.expect(jsonData).to.have.property('ProductID');",
    "  pm.expect(jsonData).to.have.property('Name');",
    "  pm.expect(jsonData).to.have.property('UnitPrice');",
    "  pm.expect(jsonData).to.have.property('StockQuantity');",
    "  pm.expect(jsonData).to.have.property('Description');",
    "});",
  ],
  },
  ],
  "request": {
    "method": "GET",
    "url": "{{base_url}}/getSingleProduct?id=1001"
  },
},
{
  "name": "Test getSingleProduct invalid ID returns 404",

```

```

"event": [
  {
    "listen": "test",
    "script": {
      "type": "text/javascript",
      "exec": [
        "pm.test('Status code is 404 for non-existent product', function () {",
        "  pm.response.to.have.status(404);",
        "});",
        "pm.test('Error detail is returned', function () {",
        "  var jsonData = pm.response.json();",
        "  pm.expect(jsonData).to.have.property('detail');",
        "});",
      ]
    }
  }
],
"request": {
  "method": "GET",
  "url": "{{base_url}}/getSingleProduct?id=99999"
},
{
  "name": "Test getAll",
  "event": [
    {
      "listen": "test",
      "script": {
        "type": "text/javascript",

```

```

"exec": [
  "pm.test('Status code is 200', function () {",
  "  pm.response.to.have.status(200);",
  "});",
  "pm.test('Response is a non-empty array', function () {",
  "  var jsonData = pm.response.json();",
  "  pm.expect(jsonData).to.be.an('array');",
  "  pm.expect(jsonData.length).to.be.above(0);",
  "});",
  "pm.test('Each product has required fields', function () {",
  "  var jsonData = pm.response.json();",
  "  jsonData.slice(0, 5).forEach(function (product) {",
  "    pm.expect(product).to.have.property('ProductID');",
  "    pm.expect(product).to.have.property('Name');",
  "    pm.expect(product).to.have.property('UnitPrice');",
  "    pm.expect(product).to.have.property('StockQuantity');",
  "    pm.expect(product).to.have.property('Description');",
  "  });",
  "});",
]
}
},
],
"request": {
  "method": "GET",
  "url": "{{base_url}}/getAll"
}
},
{

```

```

"name": "Test addNew",
"event": [
{
  "listen": "test",
  "script": {
    "type": "text/javascript",
    "exec": [
      "pm.test('Status code is 200', function () {",
      "  pm.response.to.have.status(200);",
      "});",
      "pm.test('Success message returned', function () {",
      "  var jsonData = pm.response.json();",
      "  pm.expect(jsonData.message).to.eql('Product added successfully');",
      "  pm.expect(jsonData.ProductID).to.eql(9999);",
      "});"
    ]
  }
}
],
"request": {
  "method": "POST",
  "url": "{{base_url}}/addNew",
  "header": [{"key": "Content-Type", "value": "application/json"}],
  "body": {
    "mode": "raw",
    "raw": "{\\"ProductID\\": 9999, \\"Name\\": \\"Test Product\\", \\"UnitPrice\\": 99.99, \\"StockQuantity\\": 10, \\"Description\\": \\"Test description for unit testing\\"}"
  }
}
},

```



```

{
  "name": "Test addNew duplicate returns 400",
  "event": [
    {
      "listen": "test",
      "script": {
        "type": "text/javascript",
        "exec": [
          "pm.test('Status code is 400 for duplicate ID', function () {",
          "  pm.response.to.have.status(400);",
          "});",
          "pm.test('Error detail mentions already exists', function () {",
          "  var jsonData = pm.response.json();",
          "  pm.expect(jsonData.detail).to.include('already exists');",
          "});",
          ]
        }
      }
    ],
    "request": {
      "method": "POST",
      "url": "{{base_url}}/addNew",
      "header": [{"key": "Content-Type", "value": "application/json"}],
      "body": {
        "mode": "raw",
        "raw": "{\\\"ProductID\\\": 9999, \\\"Name\\\": \\\"Duplicate Product\\\", \\\"UnitPrice\\\": 10.00, \\\"StockQuantity\\\": 1, \\\"Description\\\": \\\"Should fail as duplicate\\\"}"
      }
    }
  ],
},

```

```

{
  "name": "Test deleteOne",
  "event": [
    {
      "listen": "test",
      "script": {
        "type": "text/javascript",
        "exec": [
          "pm.test('Status code is 200', function () {",
          "  pm.response.to.have.status(200);",
          "});",
          "pm.test('Delete success message returned', function () {",
          "  var jsonData = pm.response.json();",
          "  pm.expect(jsonData.message).to.eql('Product deleted successfully');",
          "});",
        ]
      }
    }
  ],
  "request": {
    "method": "DELETE",
    "url": "{{base_url}}/deleteOne?id=9999"
  }
},
{
  "name": "Test deleteOne non-existent returns 404",
  "event": [
    {
      "listen": "test",

```

```
"script": {
  "type": "text/javascript",
  "exec": [
    "pm.test('Status code is 404 for already deleted product', function () {",
    "  pm.response.to.have.status(404);",
    "});",
  ]
}
},
{
  "request": {
    "method": "DELETE",
    "url": "{{base_url}}/deleteOne?id=9999"
  }
},
{
  "name": "Test startsWith",
  "event": [
    {
      "listen": "test",
      "script": {
        "type": "text/javascript",
        "exec": [
          "pm.test('Status code is 200', function () {",
          "  pm.response.to.have.status(200);",
          "});",
          "pm.test('Response is an array', function () {",
          "  var jsonData = pm.response.json();",
          "  pm.expect(jsonData).to.be.an('array');",
          "});"
```

```

    });",
    "pm.test('All returned products start with S', function () {",
    "    var jsonData = pm.response.json();",
    "    jsonData.forEach(function (product) {",
    "        pm.expect(product.Name.charAt(0).toUpperCase()).toEqual('S');",
    "    });",
    "});"
]
}
}
],
"request": {
    "method": "GET",
    "url": "{{base_url}}/startsWith?letter=s"
},
{
    "name": "Test paginate",
    "event": [
        {
            "listen": "test",
            "script": {
                "type": "text/javascript",
                "exec": [
                    "pm.test('Status code is 200', function () {",
                    "    pm.response.to.have.status(200);",
                    "});",
                    "pm.test('Response is an array', function () {",
                    "    var jsonData = pm.response.json();",

```

```

    "    pm.expect(jsonData).to.be.an('array');"
    "});",
    "pm.test('Returns at most 10 products', function () {",
    "    var jsonData = pm.response.json();",
    "    pm.expect(jsonData.length).to.be.at.most(10);",
    "});",
    "pm.test('Products are within the requested ID range', function () {",
    "    var jsonData = pm.response.json();",
    "    jsonData.forEach(function (product) {",
    "        pm.expect(product.ProductID).to.be.at.least(1001);",
    "        pm.expect(product.ProductID).to.be.at.most(1020);",
    "    });",
    "});",
    ]
    }
    }
],
"request": {
    "method": "GET",
    "url": "{{base_url}}/paginate?startId=1001&endId=1020"
}
},
{
    "name": "Test convert",
    "event": [
        {
            "listen": "test",
            "script": {
                "type": "text/javascript",

```

```

"exec": [
  "pm.test('Status code is 200', function () {",
    "  pm.response.to.have.status(200);",
  "});",
  "pm.test('Response contains USD and EUR prices', function () {",
    "  var jsonData = pm.response.json();",
    "  pm.expect(jsonData).to.have.property('PriceUSD');",
    "  pm.expect(jsonData).to.have.property('PriceEUR');",
    "  pm.expect(jsonData).to.have.property('ExchangeRate');",
  "});",
  "pm.test('EUR price is less than USD price', function () {",
    "  var jsonData = pm.response.json();",
    "  pm.expect(jsonData.PriceEUR).to.be.below(jsonData.PriceUSD);",
  "});",
  "pm.test('Exchange rate is a positive number', function () {",
    "  var jsonData = pm.response.json();",
    "  pm.expect(jsonData.ExchangeRate).to.be.above(0);",
  "});",
],
},
],
"request": {
  "method": "GET",
  "url": "{{base_url}}/convert?id=1001"
},
{
  "name": "Test metrics endpoint",

```

```
"event": [  
  {  
    "listen": "test",  
    "script": {  
      "type": "text/javascript",  
      "exec": [  
        "pm.test('Status code is 200', function () {",  
        "  pm.response.to.have.status(200);",  
        "});",  
        "pm.test('Response contains Prometheus metrics', function () {",  
        "  pm.expect(pm.response.text()).to.include('api_requests_total');",  
        "  pm.expect(pm.response.text()).to.include('api_request_duration_seconds');",  
        "});",  
      ],  
    },  
  },  
],  
"request": {  
  "method": "GET",  
  "url": "{{base_url}}/metrics"  
},  
],  
"variable": [  
  {"key": "base_url", "value": "http://localhost:8000"}  
]  
}
```

Docker code

Dockerfile:

```
FROM ubuntu:22.04
```

```
ENV DEBIAN_FRONTEND=noninteractive
```

```
RUN apt-get update && apt-get install -y \
```

```
python3 \
```

```
python3-pip \
```

```
python3-venv \
```

```
git \
```

```
&& rm -rf /var/lib/apt/lists/*
```

```
WORKDIR /app
```

```
# Copy application files
```

```
COPY requirements.txt .
```

```
RUN pip3 install --no-cache-dir -r requirements.txt
```

```
COPY main.py .
```

```
COPY products.csv .
```

```
# Import script for initial data load
```

```
COPY import_csv_to_mongodb.py .
```

```
EXPOSE 8000
```

```
CMD ["python3", "-m", "uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

[docker-compose.yml](#)

services:

mongodb:

image: mongo:7

volumes:

- mongodb_data:/data/db

api:

build: .

ports:

- "\${API_HOST_PORT:-8000}:8000"

environment:

- MONGODB_URI=mongodb://mongodb:27017

depends_on:

- mongodb

command: >

```
sh -c "python3 import_csv_to_mongodb.py &&  
python3 -m uvicorn main:app --host 0.0.0.0 --port 8000"
```

prometheus:

profiles: ["monitoring"]

image: prom/prometheus:latest

ports:

- "\${PROMETHEUS_HOST_PORT:-9090}:9090"

volumes:

- ./prometheus.yml:/etc/prometheus/prometheus.yml

command:

- '--config.file=/etc/prometheus/prometheus.yml'
- '--web.enable-lifecycle'

depends_on:

- api

grafana:

profiles: ["monitoring"]

image: grafana/grafana:latest

ports:

- "\${GRAFANA_HOST_PORT:-3000}:3000"

environment:

- GF_SECURITY_ADMIN_PASSWORD=admin

- GF_USERS_ALLOW_SIGN_UP=false

depends_on:

- prometheus

volumes:

mongodb_data:

prometheus.yml

Prometheus config for API monitoring

global:

scrape_interval: 15s

scrape_configs:

- job_name: 'inventory-api'

static_configs:

- targets: ['api:8000']

metrics_path: '/metrics'